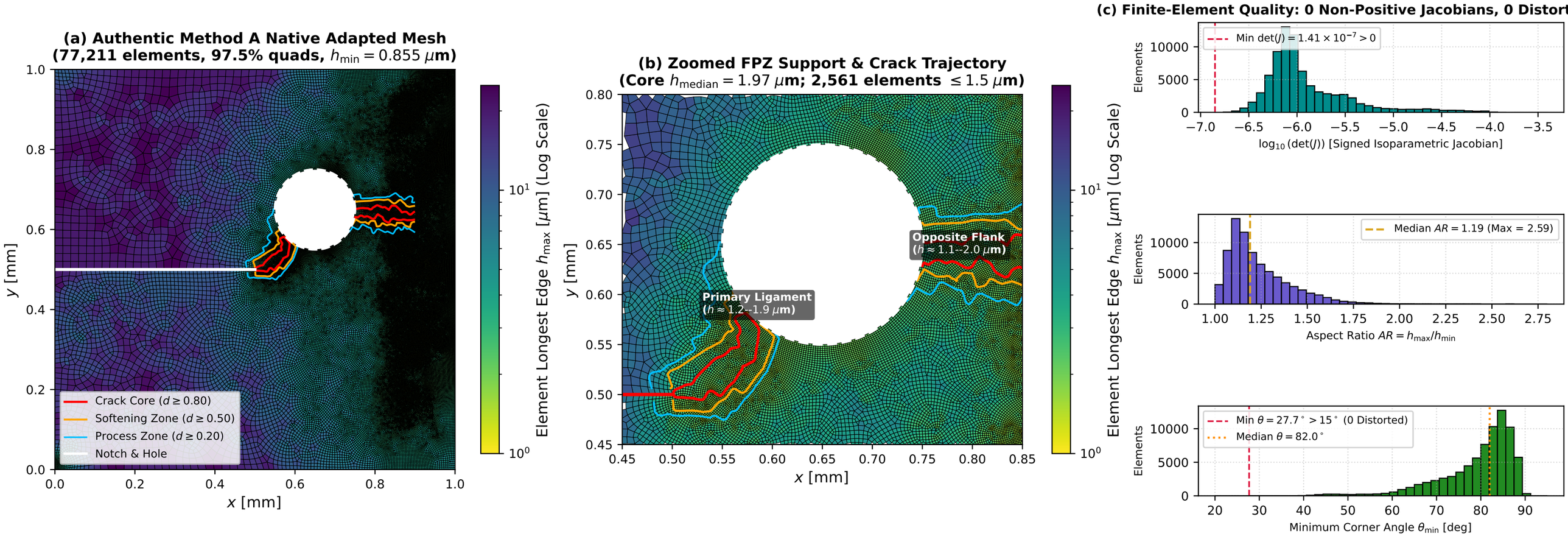

(a) Authentic Method A Native Adapted Mesh
(77,211 elements, 97.5% quads, $h_{min} = 0.855\ \mu$m)
Crack Core ($d \geq 0.80$)
Softening Zone ($d \geq 0.50$)
Process Zone ($d \geq 0.20$)
Notch & Hole
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)
(b) Zoomed FPZ Support & Crack Trajectory
(Core $h_{median} = 1.97\ \mu$m; 2,561 elements $\leq 1.5\ \mu$m)
Primary Ligament
($h \approx 1.2$–$1.9\ \mu$m)
Opposite Flank
($h \approx 1.1$–$2.0\ \mu$m)
(c) Finite-Element Quality: 0 Non-Positive Jacobians, 0 Distor
Min det(J) = $1.41 \times 10^{-7} > 0$
$\log_{10}(\det(J))$ [Signed Isoparametric Jacobian]
Median $AR = 1.19$ (Max = 2.59)
Aspect Ratio $AR = h_{max}/h_{min}$
Min $\theta = 27.7^\circ > 15^\circ$ (0 Distorted)
Median $\theta = 82.0^\circ$
Minimum Corner Angle θ_{min} [deg]
Elements